

MODELING INTERFACE EVOLUTION THROUGH TOPOLOGY CHANGES



DIFFUSE-INTERFACE REGULARIZATION

Filament collapse is regularized
by a diffuse layer of width ϵ



Quasi-incompressible Cahn–Hilliard fluids and topological transitions

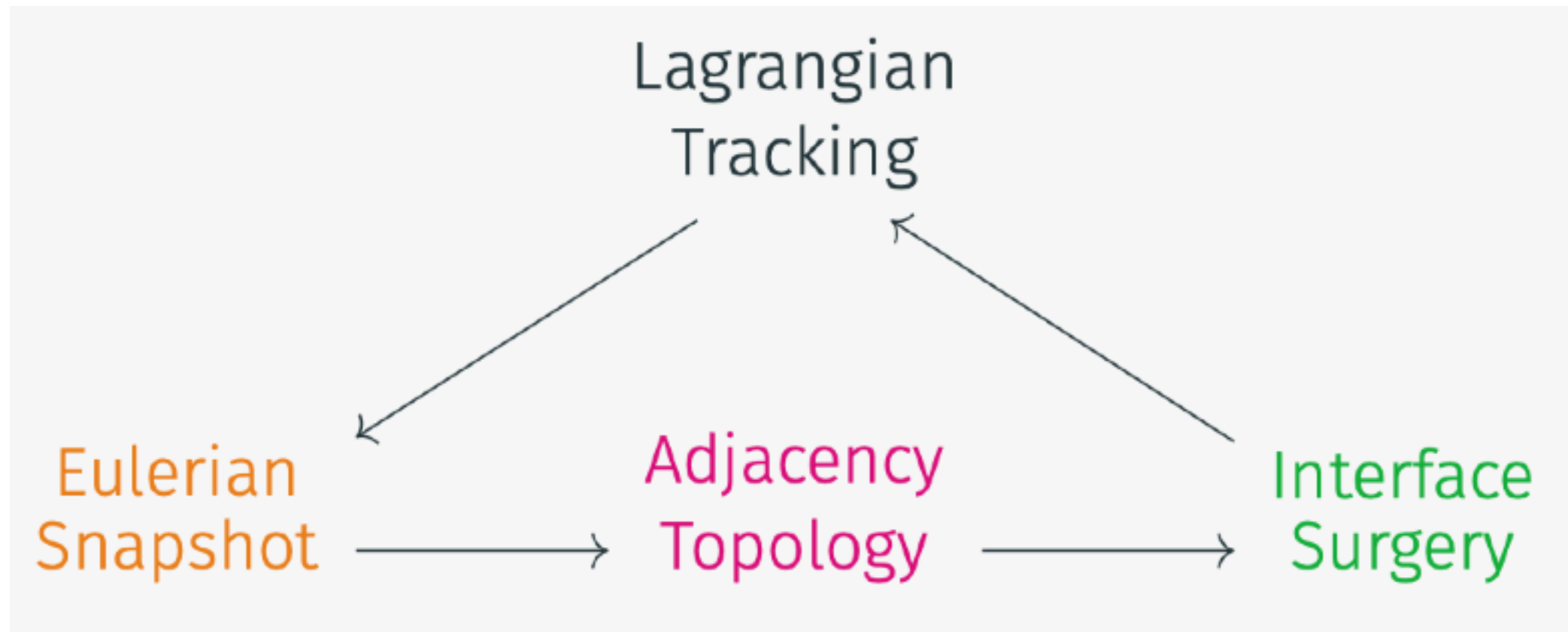
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Received 6 January 1997; revised 19 August 1997; accepted 20 November 1997

"...it is natural to conclude that the ideal numerical approach should be based on the sharp interface model and then use 'jump conditions' to transit the sharp interface through a topology change."





Sharp Interface $\longrightarrow$ **Diffuse Layer**

Topological
singularity

Regularized
transition



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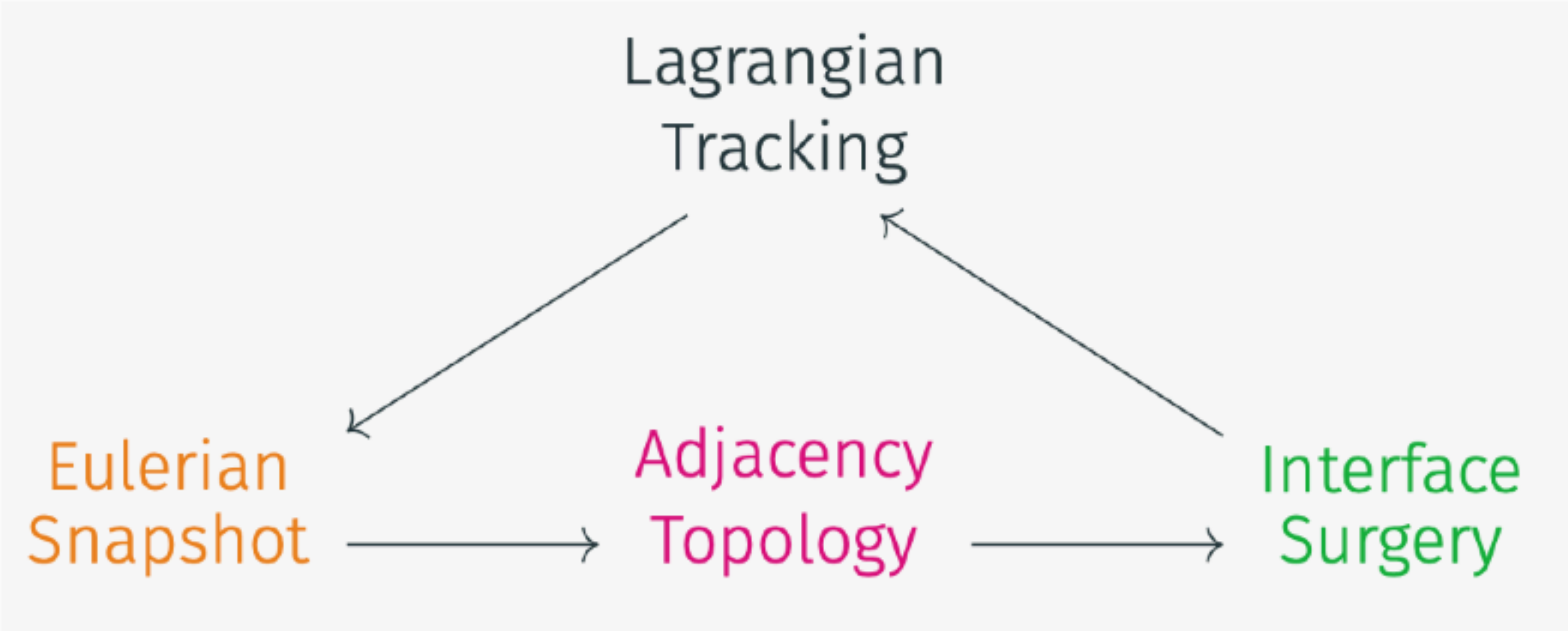
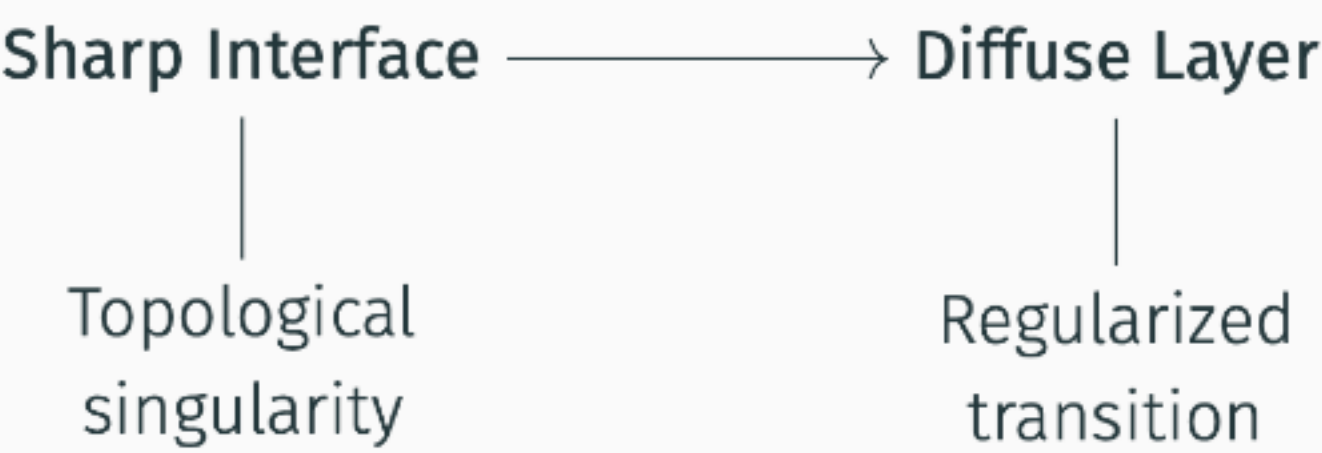
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MICROSCALE TOPOLOGICAL SURGERY (MTS)

INTERMITTENT TOPOLOGICAL TRANSITIONS VIA GEOMETRIC REGULARIZATION AT MICROSCALE

